

# AREZOO RAJABI

[rajabia@uw.edu](mailto:rajabia@uw.edu) ♦ [linkedin](#) ♦ [Website](#)♦

## AI/ML Engineer

specializing in LLM evaluation, multimodal AI, trustworthy ML, and production ML systems. 10+ years of experience developing machine learning solutions and 3+ years of experience building and evaluating LLM/GenAI systems. Proven track record delivering production AI applications serving 27,000+ users, building automated evaluation and benchmarking pipelines, optimizing multimodal inference, and developing robust ML systems. PhD in Computer Science with research and publications in AI safety, adversarial machine learning, privacy, and trustworthy AI.

## AREAS OF EXPERTISE

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LLM Evaluation & Benchmarking, Multimodal AI, Generative AI, RAG, Prompt Engineering, Agentic AI, Model Evaluation, AI Safety, Trustworthy AI, Adversarial ML, Production ML Systems

## SKILLS

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|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming:</b>                      | Python, SQL, R, MATLAB                                                                                                                        |
| <b>LLM / Generative AI:</b>              | LLM Evaluation, Benchmarking, Prompt Engineering, RAG, NLP, Tokenization, Embeddings, Transformers, Fine-Tuning, PyTorch                      |
| <b>AI Safety / Trustworthy AI:</b>       | Adversarial ML, Backdoor/Trojan Detection, Membership Inference, Privacy-Preserving ML, Federated Learning, Robustness Evaluation             |
| <b>Evaluation &amp; Experimentation:</b> | LLM-as-a-judge (RAGAS, DeepEval), Human-in-the-Loop Evaluation, Error Analysis, A/B Testing, KPI Design, Model Monitoring, Regression Testing |
| <b>MLOps &amp; Engineering:</b>          | MLflow, Docker, Git/GitHub, CI/CD, GitHub Actions                                                                                             |
| <b>Cloud &amp; Data:</b>                 | GCP, PySpark, Pandas, Hadoop, H2O                                                                                                             |
| <b>Machine Learning:</b>                 | Statistical Modeling, Experimental Design, Classification, Regression, Decision Trees, SVM, Clustering, Gradient Boosting                     |

## EXPERIENCE

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### Enterprise LLM Chatbot – 27,000+ users, \$6M ROI

- Built automated LLM evaluation and benchmarking pipelines to measure response quality, relevance, factuality, and hallucination rates across production workloads.
- Designed curated benchmark datasets and evaluation workflows to systematically compare model and prompt performance and identify failure modes.
- Developed prompt optimization and experimentation frameworks that improved response quality and reduced hallucinations across enterprise LLM workflows.
- Implemented offline and human-in-the-loop evaluation frameworks to identify model weaknesses and guide iterative improvements.
- Designed online and offline quality metrics and monitoring frameworks to track model performance and detect quality degradation in production.

### Multimodal AI for Suspicious Behavior & Policy Violation Detection – \$1.5M+ ROI

- Led development of a production multimodal AI system using Vision-Language Models for enterprise safety and policy-violation detection.
- Designed scalable video data collection, annotation, segmentation, and preprocessing pipelines supporting multimodal model evaluation and deployment.
- Reduced multimodal inference and token costs by **80%** through intelligent frame selection and automated video segmentation while preserving detection performance.

- Developed context-aware metadata and prompting strategies to improve model grounding, detection precision, and robustness.

## Applied Machine Learning

- Built predictive ML models for fraud detection, anomaly detection, and next-day demand forecasting.
- Performed feature engineering, model evaluation, error analysis, and threshold optimization to improve model performance and reduce false alarms.

## Postdoctoral Scholar

NSL Lab, University of Washington

Feb 2021 - Dec 2022

Seattle, WA, USA

- Multi-Domain Trojan Detection: Proposed a multi-domain Trojan detection framework across text, image, and audio models, achieving 85% detection accuracy through extensive benchmarking under adversarial settings.
- Privacy-Preserving RL Algorithm: Developed and evaluated privacy-preserving reinforcement learning algorithms, conducting extensive experiments to quantify privacy-utility tradeoffs under adversarial settings.
- Federated Learning for Trojan Prevention: Proposed a federated learning approach to counteract Trojan samples during the training phase.

## EDUCATION

**Ph.D. in Computer Science**, Oregon State University 2014 - 2021

**Master of Computer Engineering**, Sharif University of Technology 2011 - 2013

**Bachelor of Computer Science**, Sharif University of Technology 2005 - 2010

## SELECTED PUBLICATION

1. **A. Rajabi**, S. Asokraj, F. Jiang, L. Niu, B. Ramasubramanian, J. Ritcey, R. Poovendran, MDTD: A Multi-Domain Trojan Detector for Deep Neural Networks, ACM Conference on Computer and Communications Security (ACM CCS), Sep. 2023.
2. J. Jia, Z. Yuan, D. Sahabandu, L. Niu, **A. Rajabi**, B. Ramasubramanian, B. Li, R. Poovendran, FLGAME: A Game-theoretic Defense against Backdoor Attacks In Federated Learning, Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS), Sep 2023 (<https://neurips.cc/virtual/2023/poster/70499>).
3. **A. Rajabi**, D. Sahabandu, L. Niu, B. Ramasubramanian, R. Poovendran, LDL: A Defense for Label-Based Membership Inference Attacks, ACM Asia Conference on Computer and Communications Security (AsiaCCS), July 2023 (7% acceptance rate).
4. **A. Rajabi**, B. Ramasubramanian, A. Marruf, R. Poovendran, Privacy Preserving Reinforcement Learning Beyond Expectation, Accepted in 61st IEEE Conference on Decision and Control, 2022.
5. **A. Rajabi**, M. Abbasi, R. B. Bobba, K. Tajik, Adversarial Images Against Super-Resolution Convolutional Neural Networks for Free, Privacy Enhancing Technology Symposium (PETS), 2022.
6. M. Abbasi, **A. Rajabi**, C. Shui, C. Gagné, R. B. Bobba, Toward Adversarial Robustness by Diversity in an Ensemble of Specialized Deep Neural Networks, Canadian Conf on Artificial Intelligence, 2020. (Best Paper Award)

## PATENT

Arezo Rajabi, Dinuka Sahabandu, Luyao Niu, Bhaskar Ramasubramanian, Radha Poovendran, *LDL: A Defense for Label-Based Membership Inference Attacks*, Record of Innovation filed with CoMotion At University of Washington, Seattle, Dec. 2022.

## PROFESSIONAL SERVICES

- Adjunct Faculty at University of Washington 2022- 2024
- Organizer at The Trojan Detection Challenge (LLM Edition), [NeurIPS](#) 2023
- Organizer at Trojan Detection Challenge, [NeurIPS](#) 2022